

stores it in the cloud for future reference. The cloud keeps track of every driver's safe driving records. Every driver earns a score based on his or her driving skills, much like a credit score. The data is then stored in the cloud along with the details of when the driver violated rules, or how consistently safe he/she has been driving. "Our systems and processes have been built keeping in mind security and privacy as the fundamental design requirements," added Vinay Rai. The dependence on the cloud may give rise to questions regarding what happens when there is poor internet connectivity. The enthusiastic team at Netradyne assured that while there may be a temporary disconnectivity, there will be no change in the solution's performance. "The in-cab alert to the driver does not depend on internet connectivity. In case of poor connectivity, the alerts are stored in the device and uploaded to the cloud when the connectivity is available. The safety manager still gets the alert as and when the internet is restored. Thus, the device doesn't depend on internet connectivity for any of its processing," confirmed Vinay. As discussed in the Zoomcar case study, a significant and corresponding increase in safety can be observed in vehicles installed with Driveri with the camera feature turned on. The team at Netradyne doesn't stop here. It continues to gather more data to constantly keep track and improve the device performance. Sreekanth Annapureddy, Senior Vice President of Engineering, Analytics at Netradyne gave an insight

into the impact that Driveri has made in the fleet industry. Driveri analyses data from multiple camera sources and inertial sensors in real-time to make sense of the traffic situation and driving behaviours. For instance, with the advanced AI technology being able to assess and alert distracted driving behaviours instantly, there has been a whopping 60% reduction in distracted driving, 76% improvement in maintaining distance between vehicles and a 90% better compliance score. With the help of Driveri, customers have seen nearly 10 times reduction in risky driving behaviour due to host of reasons such as talking or texting on phone while driving, not wearing seatbelt, exceeding speed limit, jumping traffic light when it is red etc. Sreekanth stated that the company has been processing over 1 trillion sensor data points every month, which puts into perspective the scale of operations at Netradyne. In essence, the engineering team trains Machine Learning (ML) models to understand the various aspects of driving. Each aspect has its own long-tail corner case scenarios to study and analyse. For eg. the scenario of driver looking down is classified as distracted driving whereas the same scenario of a driver's head tilted down but with eyes on the road is classified as attentive driving. "Netradyne's unique business model allows us to accumulate large datasets comprising various corner cases that power our AI engine," Sreekanth explains. "We not only optimise the ML models for real-time inference

on multiple compute environments but also train it constantly to understand the traffic rules across various countries. With the ever-growing demand to cram more functionality into the Driveri devices, we are constantly pushing the limits of what's possible in applied AI research," he adds.

The dynamic team at Netradyne is united by passion unlike any other Indian MNC with a visionary leadership team and global outlook. The team comprises like-minded people that have a thirst for knowledge and a pursuit for perfection in all that they do. Coming from institutions that instil values of dedication and hard work, the team is now hoping to expand their circle with people who share the passion, vision and value systems.

Netradyne invites ML Engineers, Data Scientists, Software Engineers who are passionate about solving real-world problems and working on modules end-to-end. Netradyne provides a vibrant place for young talent to interact with world-class engineers and learn various aspects of Hardware Design, Software Design Patterns, Deep Learning, Statistics, MLOPs, Databases, Scalable Micro-services, Automated Testing and Operational Excellence.

If you think you have the skillset to contribute to the knowledgeable team that manufactures a high functioning, life-saving product in India for the world, then be sure to head to the careers page of

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